
Diophantine approximation

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The diophantine approximation deals with the approximation of real numbers (or real vectors) with rational numbers (or rational vectors). See the article [Wikipedia article Diophantine_approximation](#) for more information.

CONTINUED FRACTIONS

A continued fraction is a representation of a real number in terms of a sequence of integers denoted $[a_0; a_1, a_2, \dots]$. The well known decimal expansion is another way of representing a real number by a sequence of integers. The value of a continued fraction is defined recursively as:

$$[a_0; a_1, a_2, \dots] = a_0 + \frac{1}{[a_1; a_2, \dots]} = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\dots}}}$$

In this expansion, all coefficients a_n are integers and only the value a_0 may be non positive. Note that a_0 is nothing else but the floor (this remark provides a way to build the continued fraction expansion from a given real number). As examples

$$\frac{45}{38} = 1 + \frac{1}{5 + \frac{1}{2 + \frac{1}{3}}}$$
$$\pi = 3 + \frac{1}{7 + \frac{1}{15 + \frac{1}{1 + \frac{1}{292 + \frac{1}{\dots}}}}}$$

It is quite remarkable that

- any real number admits a unique continued fraction expansion
- finite expansions correspond to rationals
- ultimately periodic expansions correspond to quadratic numbers (ie numbers of the form $a + b\sqrt{D}$ with a and b rationals and D square free positive integer)
- two real numbers x and y have the same tail (up to a shift) in their continued fraction expansion if and only if there are integers a, b, c, d with $|ad - bc| = 1$ and such that $y = (ax + b)/(cx + d)$.

Moreover, the rational numbers obtained by truncation of the expansion of a real number gives its so-called best approximations. For more informations on continued fractions, you may have a look at [Wikipedia article Continued_fraction](#).

EXAMPLES:

If you want to create the continued fraction of some real number you may either use its method `continued_fraction` (if it exists) or call `continued_fraction()`:

```
sage: (13/27).continued_fraction()
[0; 2, 13]
sage: 0 + 1/(2 + 1/13)
13/27

sage: continued_fraction(22/45)
[0; 2, 22]
sage: 0 + 1/(2 + 1/22)
22/45

sage: continued_fraction(pi)                                     #_
↳needs sage.symbolic
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: continued_fraction_list(pi, nterms=5)                     #_
↳needs sage.symbolic
[3, 7, 15, 1, 292]

sage: x = polygen(ZZ, 'x')
sage: K.<cbt5> = NumberField(x^3 - 5, embedding=1.709)             #_
↳needs sage.rings.number_field
sage: continued_fraction(cbt5)                                   #_
↳needs sage.rings.number_field
[1; 1, 2, 2, 4, 3, 3, 1, 5, 1, 1, 4, 10, 17, 1, 14, 1, 1, 3052, 1, ...]
```

```
>>> from sage.all import *
>>> (Integer(13)/Integer(27)).continued_fraction()
[0; 2, 13]
>>> Integer(0) + Integer(1)/(Integer(2) + Integer(1)/Integer(13))
13/27

>>> continued_fraction(Integer(22)/Integer(45))
[0; 2, 22]
>>> Integer(0) + Integer(1)/(Integer(2) + Integer(1)/Integer(22))
22/45

>>> continued_fraction(pi)                                     #_
↳needs sage.symbolic
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> continued_fraction_list(pi, nterms=Integer(5))             #_
↳ # needs sage.symbolic
[3, 7, 15, 1, 292]

>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(5), embedding=RealNumber('1.709'), names=(
↳'cbt5',)); (cbt5,) = K._first_ngens(1) # needs sage.rings.number_field
>>> continued_fraction(cbt5)                                   #_
↳needs sage.rings.number_field
[1; 1, 2, 2, 4, 3, 3, 1, 5, 1, 1, 4, 10, 17, 1, 14, 1, 1, 3052, 1, ...]
```

It is also possible to create a continued fraction from a list of partial quotients:

```
sage: continued_fraction([-3,1,2,3,4,1,2])
[-3; 1, 2, 3, 4, 1, 2]
```

```
>>> from sage.all import *
>>> continued_fraction([-Integer(3), Integer(1), Integer(2), Integer(3), Integer(4),
↳Integer(1), Integer(2)])
```

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```
[-3; 1, 2, 3, 4, 1, 2]
```

Even infinite:

```
sage: w = words.ThueMorseWord([1,2]); w #_
↪needs sage.combinat
word: 1221211221121221211212211221211221121221...
sage: continued_fraction(w) #_
↪needs sage.combinat
[1; 2, 2, 1, 2, 1, 1, 2, 2, 1...]
```

```
>>> from sage.all import *
>>> w = words.ThueMorseWord([Integer(1),Integer(2)]); w #_
↪ # needs sage.combinat
word: 1221211221121221211212211221211221121221...
>>> continued_fraction(w) #_
↪needs sage.combinat
[1; 2, 2, 1, 2, 1, 1, 2, 2, 1...]
```

To go back and forth between the value (as a real number) and the partial quotients (seen as a finite or infinite list) you can use the methods `quotients` and `value`:

```
sage: cf = (13/27).continued_fraction()
sage: cf.quotients()
[0, 2, 13]
sage: cf.value()
13/27

sage: cf = continued_fraction(pi) #_
↪needs sage.symbolic
sage: cf.quotients() #_
↪needs sage.symbolic
lazy list [3, 7, 15, ...]
sage: cf.value() #_
↪needs sage.symbolic
pi

sage: # needs sage.combinat
sage: w = words.FibonacciWord([1,2])
sage: cf = continued_fraction(w)
sage: cf.quotients()
word: 1211212112112121121121121121121121121...
sage: v = cf.value(); v
1.387954587967143?
sage: v.n(digits=100)
1.
↪38795458796714233691931385987318547787815245249853227189491728982641857762264893216988523703424296...
sage: v.continued_fraction()
[1; 2, 1, 1, 2, 1, 2, 1, 1, 2, 1, 1, 2, 1, 2, 1, 1, 2, 1, 2...]
```

```
>>> from sage.all import *
>>> cf = (Integer(13)/Integer(27)).continued_fraction()
>>> cf.quotients()
[0, 2, 13]
>>> cf.value()
13/27
```

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```
>>> cf = continued_fraction(pi)                                     #_
↳needs sage.symbolic
>>> cf.quotients()                                                 #_
↳needs sage.symbolic
lazy list [3, 7, 15, ...]
>>> cf.value()                                                     #_
↳needs sage.symbolic
pi

>>> # needs sage.combinat
>>> w = words.FibonacciWord([Integer(1), Integer(2)])
>>> cf = continued_fraction(w)
>>> cf.quotients()
word: 1211212112112121121121121121121121121121121121121...
>>> v = cf.value(); v
1.387954587967143?
>>> v.n(digits=Integer(100))
1.
↳38795458796714233691931385987318547787815245249853227189491728982641857762264893216988523703424296...
>>> v.continued_fraction()
[1; 2, 1, 1, 2, 1, 2, 1, 1, 2, 1, 1, 2, 1, 1, 2, 1, 2, 1, 1, 2, 1, 2...]
```

Recall that quadratic numbers correspond to ultimately periodic continued fractions. For them special methods give access to preperiod and period:

```
sage: # needs sage.rings.number_field
sage: K.<sqrt2> = QuadraticField(2)
sage: cf = continued_fraction(sqrt2); cf
[1; (2)*]
sage: cf.value()
sqrt2
sage: cf.preperiod()
(1,)
sage: cf.period()
(2,)

sage: cf = (3*sqrt2 + 1/2).continued_fraction(); cf               #_
↳needs sage.rings.number_field
[4; (1, 2, 1, 7)*]

sage: cf = continued_fraction([(1,2,3), (1,4)]); cf
[1; 2, 3, (1, 4)*]
sage: cf.value()                                                  #_
↳needs sage.rings.number_field
-2/23*sqrt2 + 36/23
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = QuadraticField(Integer(2), names=('sqrt2',)); (sqrt2,) = K._first_ngens(1)
>>> cf = continued_fraction(sqrt2); cf
[1; (2)*]
>>> cf.value()
sqrt2
>>> cf.preperiod()
(1,)
```

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```
>>> cf.period()
(2,)

>>> cf = (Integer(3)*sqrt2 + Integer(1)/Integer(2)).continued_fraction(); cf
↳ # needs sage.rings.number_field
[4; (1, 2, 1, 7)*]

>>> cf = continued_fraction([(Integer(1),Integer(2),Integer(3)), (Integer(1),
↳ Integer(4))]); cf
[1; 2, 3, (1, 4)*]
>>> cf.value()
↳ needs sage.rings.number_field
-2/23*sqrt2 + 36/23
```

On the following we can remark how the tail may change even in the same quadratic field:

```
sage: for i in range(20): print(continued_fraction(i*sqrt2))
↳ needs sage.rings.number_field
[0]
[1; (2)*]
[2; (1, 4)*]
[4; (4, 8)*]
[5; (1, 1, 1, 10)*]
[7; (14)*]
...
[24; (24, 48)*]
[25; (2, 5, 6, 5, 2, 50)*]
[26; (1, 6, 1, 2, 3, 2, 26, 2, 3, 2, 1, 6, 1, 52)*]
```

```
>>> from sage.all import *
>>> for i in range(Integer(20)): print(continued_fraction(i*sqrt2))
↳ # needs sage.rings.number_field
[0]
[1; (2)*]
[2; (1, 4)*]
[4; (4, 8)*]
[5; (1, 1, 1, 10)*]
[7; (14)*]
...
[24; (24, 48)*]
[25; (2, 5, 6, 5, 2, 50)*]
[26; (1, 6, 1, 2, 3, 2, 26, 2, 3, 2, 1, 6, 1, 52)*]
```

Nevertheless, the tail is preserved under invertible integer homographies:

```
sage: # needs sage.modular sage.rings.number_field
sage: apply_homography = lambda m,z: (m[0,0]*z + m[0,1]) / (m[1,0]*z + m[1,1])
sage: m1 = SL2Z([60,13,83,18])
sage: m2 = SL2Z([27,80,28,83])
sage: a = sqrt2/3
sage: a.continued_fraction()
[0; 2, (8, 4)*]
sage: b = apply_homography(m1, a)
sage: b.continued_fraction()
[0; 1, 2, 1, 1, 1, 1, 6, (8, 4)*]
sage: c = apply_homography(m2, a)
```

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```
sage: c.continued_fraction()
[0; 1, 26, 1, 2, 2, (8, 4)*]
sage: d = apply_homography(m1**2*m2**3, a)
sage: d.continued_fraction()
[0; 1, 2, 1, 1, 1, 1, 5, 2, 1, 1, 1, 1, 5, 26, 1, 2, 1, 26, 1, 2, 1, 26, 1, 2, 2, (8, 4)*]
```

```
>>> from sage.all import *
>>> # needs sage.modular sage.rings.number_field
>>> apply_homography = lambda m,z: (m[Integer(0),Integer(0)]*z + m[Integer(0),
↳ Integer(1)]) / (m[Integer(1),Integer(0)]*z + m[Integer(1),Integer(1)])
>>> m1 = SL2Z([Integer(60),Integer(13),Integer(83),Integer(18)])
>>> m2 = SL2Z([Integer(27),Integer(80),Integer(28),Integer(83)])
>>> a = sqrt2/Integer(3)
>>> a.continued_fraction()
[0; 2, (8, 4)*]
>>> b = apply_homography(m1, a)
>>> b.continued_fraction()
[0; 1, 2, 1, 1, 1, 1, 6, (8, 4)*]
>>> c = apply_homography(m2, a)
>>> c.continued_fraction()
[0; 1, 26, 1, 2, 2, (8, 4)*]
>>> d = apply_homography(m1**Integer(2)*m2**Integer(3), a)
>>> d.continued_fraction()
[0; 1, 2, 1, 1, 1, 1, 5, 2, 1, 1, 1, 1, 5, 26, 1, 2, 1, 26, 1, 2, 1, 26, 1, 2, 2, (8, 4)*]
```

Todo:

- Improve numerical approximation (the method `_mpfr_()` is quite slow compared to the same method for an element of a number field)
- Make a class for generalized continued fractions of the form $a_0 + b_0/(a_1 + b_1/(...))$ (the standard continued fractions are when all $b_n = 1$ while the Hirzebruch-Jung continued fractions are the one for which $b_n = -1$ for all n). See [Wikipedia article Generalized continued fraction](#).
- look at the function `ContinuedFractionApproximationOfRoot` in GAP

AUTHORS:

- Vincent Delecroix (2014): cleaning, refactorisation, documentation from the old implementation in `contfrac` ([Issue #14567](#)).

class `sage.rings.continued_fraction.ContinuedFraction_base`

Bases: `SageObject`

Base class for (standard) continued fractions.

If you want to implement your own continued fraction, simply derived from this class and implement the following methods:

- `def quotient(self, n):` return the n -th quotient of `self` as a Sage integer
- `def length(self):` the number of partial quotients of `self` as a Sage integer or `Infinity`.

and optionally:

- `def value(self):` return the value of `self` (an exact real number)

This base class will provide:

- computation of convergents in `convergent()`, `numerator()` and `denominator()`
- comparison with other continued fractions (see `__richcmp__()`)
- elementary arithmetic function `floor()`, `ceil()`, `sign()`
- accurate numerical approximations `_mpfr_()`

All other methods, in particular the ones involving binary operations like sum or product, rely on the optional method `value()` (and not on convergents) and may fail at execution if it is not implemented.

additive_order()

Return the additive order of this continued fraction, which we defined to be the additive order of its value.

EXAMPLES:

```
sage: continued_fraction(-1).additive_order()
+Infinity
sage: continued_fraction(0).additive_order()
1
```

```
>>> from sage.all import *
>>> continued_fraction(-Integer(1)).additive_order()
+Infinity
>>> continued_fraction(Integer(0)).additive_order()
1
```

apply_homography(a, b, c, d, forward_value=False)

Return the continued fraction of $(ax + b)/(cx + d)$.

This is computed using Gosper's algorithm, see `continued_fraction_gosper`.

INPUT:

- `a, b, c, d` – integers
- `forward_value` – boolean (default: `False`) whether the returned continued fraction is given the symbolic value of $(ax + b)/(cx + d)$ and not only the list of partial quotients obtained from Gosper's algorithm.

EXAMPLES:

```
sage: (5 * 13/6 - 2) / (3 * 13/6 - 4)
53/15
sage: continued_fraction(13/6).apply_homography(5, -2, 3, -4).value()
53/15
```

```
>>> from sage.all import *
>>> (Integer(5) * Integer(13)/Integer(6) - Integer(2)) / (Integer(3) * _
↳ Integer(13)/Integer(6) - Integer(4))
53/15
>>> continued_fraction(Integer(13)/Integer(6)).apply_homography(Integer(5), -
↳ Integer(2), Integer(3), -Integer(4)).value()
53/15
```

We demonstrate now the effect of the optional argument `forward_value`:

```
sage: cf = continued_fraction(pi) #_
↪needs sage.symbolic
sage: h1 = cf.apply_homography(35, -27, 12, -5); h1 #_
↪needs sage.symbolic
[2; 1, 1, 6, 3, 1, 2, 1, 5, 3, 1, 1, 1, 1, 9, 12, 1, 1, 1, 3...]
sage: h1.value() #_
↪needs sage.symbolic
2.536941776086946?

sage: h2 = cf.apply_homography(35, -27, 12, -5, forward_value=True); h2 #_
↪needs sage.symbolic
[2; 1, 1, 6, 3, 1, 2, 1, 5, 3, 1, 1, 1, 1, 9, 12, 1, 1, 1, 3...]
sage: h2.value() #_
↪needs sage.symbolic
(35*pi - 27)/(12*pi - 5)
```

```
>>> from sage.all import *
>>> cf = continued_fraction(pi) #_
↪needs sage.symbolic
>>> h1 = cf.apply_homography(Integer(35), -Integer(27), Integer(12), -
↪Integer(5)); h1 # needs sage.symbolic
[2; 1, 1, 6, 3, 1, 2, 1, 5, 3, 1, 1, 1, 1, 9, 12, 1, 1, 1, 3...]
>>> h1.value() #_
↪needs sage.symbolic
2.536941776086946?

>>> h2 = cf.apply_homography(Integer(35), -Integer(27), Integer(12), -
↪Integer(5), forward_value=True); h2 # needs sage.symbolic
[2; 1, 1, 6, 3, 1, 2, 1, 5, 3, 1, 1, 1, 1, 9, 12, 1, 1, 1, 3...]
>>> h2.value() #_
↪needs sage.symbolic
(35*pi - 27)/(12*pi - 5)
```

REFERENCES:

- [Gos1972]
- [Knu1998] Exercise 4.5.3.15
- [LS1998]

`ceil()`

Return the ceil of self.

EXAMPLES:

```
sage: cf = continued_fraction([2,1,3,4])
sage: cf.ceil()
3
```

```
>>> from sage.all import *
>>> cf = continued_fraction([Integer(2),Integer(1),Integer(3),Integer(4)])
>>> cf.ceil()
3
```

`convergent(n)`

Return the n-th partial convergent to self.

EXAMPLES:

```

sage: a = continued_fraction(pi); a                                     #_
↳needs sage.symbolic
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: a.convergent(3)                                                #_
↳needs sage.symbolic
355/113
sage: a.convergent(15)                                               #_
↳needs sage.symbolic
411557987/131002976

```

```

>>> from sage.all import *
>>> a = continued_fraction(pi); a                                     #_
↳needs sage.symbolic
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> a.convergent(Integer(3))                                         _
↳ # needs sage.symbolic
355/113
>>> a.convergent(Integer(15))                                       _
↳ # needs sage.symbolic
411557987/131002976

```

convergents()

Return the list of partial convergents of self.

If self is an infinite continued fraction, then the object returned is a `lazy_list_generic` which behave like an infinite list.

EXAMPLES:

```

sage: a = continued_fraction(23/157); a
[0; 6, 1, 4, 1, 3]
sage: a.convergents()
[0, 1/6, 1/7, 5/34, 6/41, 23/157]

```

```

>>> from sage.all import *
>>> a = continued_fraction(Integer(23)/Integer(157)); a
[0; 6, 1, 4, 1, 3]
>>> a.convergents()
[0, 1/6, 1/7, 5/34, 6/41, 23/157]

```

Todo: Add an example with infinite list.

denominator(n)

Return the denominator of the n-th partial convergent of self.

EXAMPLES:

```

sage: # needs sage.symbolic
sage: c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: c.denominator(0)
1
sage: c.denominator(12)
25510582
sage: c.denominator(152)
1255341492699841451528811722575401081588363886480089431843026103930863337221076748

```

```
>>> from sage.all import *
>>> # needs sage.symbolic
>>> c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> c.denominator(Integer(0))
1
>>> c.denominator(Integer(12))
25510582
>>> c.denominator(Integer(152))
1255341492699841451528811722575401081588363886480089431843026103930863337221076748
```

floor()

Return the floor of self.

EXAMPLES:

```
sage: cf = continued_fraction([2,1,2,3])
sage: cf.floor()
2
```

```
>>> from sage.all import *
>>> cf = continued_fraction([Integer(2),Integer(1),Integer(2),Integer(3)])
>>> cf.floor()
2
```

is_minus_one()

Test whether self is minus one.

EXAMPLES:

```
sage: continued_fraction(-1).is_minus_one()
True
sage: continued_fraction(1).is_minus_one()
False
sage: continued_fraction(0).is_minus_one()
False
sage: continued_fraction(-2).is_minus_one()
False
sage: continued_fraction([-1,1]).is_minus_one()
False
```

```
>>> from sage.all import *
>>> continued_fraction(-Integer(1)).is_minus_one()
True
>>> continued_fraction(Integer(1)).is_minus_one()
False
>>> continued_fraction(Integer(0)).is_minus_one()
False
>>> continued_fraction(-Integer(2)).is_minus_one()
False
>>> continued_fraction([-Integer(1),Integer(1)]).is_minus_one()
False
```

is_one()

Test whether self is one.

EXAMPLES:

```
sage: continued_fraction(1).is_one()
True
sage: continued_fraction(5/4).is_one()
False
sage: continued_fraction(0).is_one()
False
sage: continued_fraction(pi).is_one()
↪needs sage.symbolic
False
```

```
>>> from sage.all import *
>>> continued_fraction(Integer(1)).is_one()
True
>>> continued_fraction(Integer(5)/Integer(4)).is_one()
False
>>> continued_fraction(Integer(0)).is_one()
False
>>> continued_fraction(pi).is_one()
↪needs sage.symbolic
False
```

is_zero()

Test whether self is zero.

EXAMPLES:

```
sage: continued_fraction(0).is_zero()
True
sage: continued_fraction((0,1)).is_zero()
False
sage: continued_fraction(-1/2).is_zero()
False
sage: continued_fraction(pi).is_zero()
↪needs sage.symbolic
False
```

```
>>> from sage.all import *
>>> continued_fraction(Integer(0)).is_zero()
True
>>> continued_fraction((Integer(0),Integer(1))).is_zero()
False
>>> continued_fraction(-Integer(1)/Integer(2)).is_zero()
False
>>> continued_fraction(pi).is_zero()
↪needs sage.symbolic
False
```

multiplicative_order()

Return the multiplicative order of this continued fraction, which we defined to be the multiplicative order of its value.

EXAMPLES:

```
sage: continued_fraction(-1).multiplicative_order()
2
sage: continued_fraction(1).multiplicative_order()
```

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```
1
sage: continued_fraction(pi).multiplicative_order() #_
↳needs sage.symbolic
+Infinity
```

```
>>> from sage.all import *
>>> continued_fraction(-Integer(1)).multiplicative_order()
2
>>> continued_fraction(Integer(1)).multiplicative_order()
1
>>> continued_fraction(pi).multiplicative_order() #_
↳needs sage.symbolic
+Infinity
```

n (*prec=None, digits=None, algorithm=None*)

Return a numerical approximation of this continued fraction with *prec* bits (or decimal *digits*) of precision.

INPUT:

- *prec* – precision in bits
- *digits* – precision in decimal digits (only used if *prec* is not given)
- *algorithm* – ignored for continued fractions

If neither *prec* nor *digits* is given, the default precision is 53 bits (roughly 16 digits).

EXAMPLES:

```
sage: w = words.FibonacciWord([1,3]) #_
↳needs sage.combinat
sage: cf = continued_fraction(w); cf #_
↳needs sage.combinat
[1; 3, 1, 1, 3, 1, 3, 1, 1, 3, 1, 1, 3, 1, 3, 1, 1, 3, 1, 3...]
sage: cf.numerical_approx(prec=53) #_
↳needs sage.combinat
1.28102513329557
```

```
>>> from sage.all import *
>>> w = words.FibonacciWord([Integer(1),Integer(3)]) #_
↳ # needs sage.combinat
>>> cf = continued_fraction(w); cf #_
↳needs sage.combinat
[1; 3, 1, 1, 3, 1, 3, 1, 1, 3, 1, 1, 3, 1, 3, 1, 1, 3, 1, 3...]
>>> cf.numerical_approx(prec=Integer(53)) #_
↳ # needs sage.combinat
1.28102513329557
```

The method *n* is a shortcut to this one:

```
sage: cf.n(digits=25) #_
↳needs sage.combinat
1.281025133295569815552930
sage: cf.n(digits=33) #_
↳needs sage.combinat
1.28102513329556981555293038097590
```

```
>>> from sage.all import *
>>> cf.n(digits=Integer(25))
↪ # needs sage.combinat
1.281025133295569815552930
>>> cf.n(digits=Integer(33))
↪ # needs sage.combinat
1.28102513329556981555293038097590
```

numerator (*n*)

Return the numerator of the *n*-th partial convergent of self.

EXAMPLES:

```
sage: # needs sage.symbolic
sage: c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: c.numerator(0)
3
sage: c.numerator(12)
80143857
sage: c.numerator(152)
39437716112122266962743738812600748213157266596588744951727393497446921245353005283
```

```
>>> from sage.all import *
>>> # needs sage.symbolic
>>> c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> c.numerator(Integer(0))
3
>>> c.numerator(Integer(12))
80143857
>>> c.numerator(Integer(152))
39437716112122266962743738812600748213157266596588744951727393497446921245353005283
```

numerical_approx (*prec=None, digits=None, algorithm=None*)

Return a numerical approximation of this continued fraction with *prec* bits (or decimal digits) of precision.

INPUT:

- *prec* – precision in bits
- *digits* – precision in decimal digits (only used if *prec* is not given)
- *algorithm* – ignored for continued fractions

If neither *prec* nor *digits* is given, the default precision is 53 bits (roughly 16 digits).

EXAMPLES:

```
sage: w = words.FibonacciWord([1,3])
↪ needs sage.combinat
sage: cf = continued_fraction(w); cf
↪ needs sage.combinat
[1; 3, 1, 1, 3, 1, 3, 1, 1, 3, 1, 1, 3, 1, 3, 1, 1, 3, 1, 3...]
sage: cf.numerical_approx(prec=53)
↪ needs sage.combinat
1.28102513329557
```

```
>>> from sage.all import *
>>> w = words.FibonacciWord([Integer(1), Integer(3)])
↳ # needs sage.combinat
>>> cf = continued_fraction(w); cf
↳ # needs sage.combinat
[1; 3, 1, 1, 3, 1, 3, 1, 1, 3, 1, 1, 3, 1, 3, 1, 1, 3, 1, 3...]
>>> cf.numerical_approx(prec=Integer(53))
↳ # needs sage.combinat
1.28102513329557
```

The method `n` is a shortcut to this one:

```
sage: cf.n(digits=25)
↳ # needs sage.combinat
1.281025133295569815552930
sage: cf.n(digits=33)
↳ # needs sage.combinat
1.28102513329556981555293038097590
```

```
>>> from sage.all import *
>>> cf.n(digits=Integer(25))
↳ # needs sage.combinat
1.281025133295569815552930
>>> cf.n(digits=Integer(33))
↳ # needs sage.combinat
1.28102513329556981555293038097590
```

p(*n*)

Return the numerator of the *n*-th partial convergent of `self`.

EXAMPLES:

```
sage: # needs sage.symbolic
sage: c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: c.numerator(0)
3
sage: c.numerator(12)
80143857
sage: c.numerator(152)
3943771611212266962743738812600748213157266596588744951727393497446921245353005283
```

```
>>> from sage.all import *
>>> # needs sage.symbolic
>>> c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> c.numerator(Integer(0))
3
>>> c.numerator(Integer(12))
80143857
>>> c.numerator(Integer(152))
3943771611212266962743738812600748213157266596588744951727393497446921245353005283
```

q(*n*)

Return the denominator of the *n*-th partial convergent of `self`.

EXAMPLES:

```

sage: # needs sage.symbolic
sage: c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: c.denominator(0)
1
sage: c.denominator(12)
25510582
sage: c.denominator(152)
1255341492699841451528811722575401081588363886480089431843026103930863337221076748

```

```

>>> from sage.all import *
>>> # needs sage.symbolic
>>> c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> c.denominator(Integer(0))
1
>>> c.denominator(Integer(12))
25510582
>>> c.denominator(Integer(152))
1255341492699841451528811722575401081588363886480089431843026103930863337221076748

```

quotients()

Return the list of partial quotients of `self`.

If `self` is an infinite continued fraction, then the object returned is a `lazy_list_generic` which behaves like an infinite list.

EXAMPLES:

```

sage: a = continued_fraction(23/157); a
[0; 6, 1, 4, 1, 3]
sage: a.quotients()
[0, 6, 1, 4, 1, 3]

```

```

>>> from sage.all import *
>>> a = continued_fraction(Integer(23)/Integer(157)); a
[0; 6, 1, 4, 1, 3]
>>> a.quotients()
[0, 6, 1, 4, 1, 3]

```

Todo: Add an example with infinite list.

sign()

Return the sign of `self` as an Integer.

The sign is defined to be 0 if `self` is 0, 1 if `self` is positive and -1 if `self` is negative.

EXAMPLES:

```

sage: continued_fraction(tan(pi/7)).sign()
↪needs sage.symbolic
1
sage: continued_fraction(-34/2115).sign()
-1
sage: continued_fraction([0]).sign()
0

```

```
>>> from sage.all import *
>>> continued_fraction(tan(pi/Integer(7))).sign()
↪ # needs sage.symbolic
1
>>> continued_fraction(-Integer(34)/Integer(2115)).sign()
-1
>>> continued_fraction([Integer(0)]).sign()
0
```

str (nterms=10, unicode=False, join=True)

Return a string representing this continued fraction.

INPUT:

- **nterms** – the maximum number of terms to use
- **unicode** – (default False) whether to use unicode character
- **join** – (default True) if False instead of returning a string return a list of string, each of them representing a line

EXAMPLES:

```
sage: print(continued_fraction(pi).str())
↪ # needs sage.symbolic
3 + -----
      1
      7 + -----
            1
            15 + -----
                  1
                  1 + -----
                        1
                        292 + -----
                              1
                              1 + -----
                                    1
                                    1 + -----
                                          1
                                          1 + -----
                                                1
                                                2 + -----
                                                      1
                                                      1 + ...

sage: print(continued_fraction(pi).str(nterms=1))
↪ # needs sage.symbolic
3 + ...

sage: print(continued_fraction(pi).str(nterms=2))
↪ # needs sage.symbolic
3 + -----
      1
      7 + ...

sage: print(continued_fraction(243/354).str())
-----
1
1 + -----
      1
```

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```

      2 + -----
            1
          5 + -----
                1
              3 + ---
                    2
sage: continued_fraction(243/354).str(join=False)
['      1',
 '-----',
 '      1',
 '1 + -----',
 '      1',
 '2 + -----',
 '      1',
 '5 + -----',
 '      1',
 '3 + ---',
 '      2']

sage: print(continued_fraction(243/354).str(unicode=True))
      1
-----
1 + -----
      1
    2 + -----
          1
        5 + -----
              1
            3 + -----
                  2

```

```

>>> from sage.all import *
>>> print(continued_fraction(pi).str()) #_
↳ needs sage.symbolic
      1
3 + -----
            1
          7 + -----
                1
              15 + -----
                    1
                    1 + -----
                        1
                        292 + -----
                              1
                              1 + -----
                                  1
                                  1 + -----
                                      1
                                      1 + -----
                                          1
                                          2 + -----
                                              1 + ...
>>> print(continued_fraction(pi).str(terms=Integer(1)))
↳ # needs sage.symbolic

```

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```

3 + ...
>>> print(continued_fraction(pi).str(nterms=Integer(2)))
↪ # needs sage.symbolic
      1
3 + ----
      7 + ...

>>> print(continued_fraction(Integer(243)/Integer(354)).str())
      1
-----
      1
1 + ----
      1
      2 + ----
          1
          5 + ----
              1
              3 + ----
                  2

>>> continued_fraction(Integer(243)/Integer(354)).str(join=False)
['      1',
 '-----',
 '      1',
 '1 + ----',
 '      1',
 '      2 + ----',
 '          1',
 '          5 + ----',
 '              1',
 '              3 + ----',
 '                  2']

>>> print(continued_fraction(Integer(243)/Integer(354)).str(unicode=True))
      1
-----
      1
1 + ----
      1
      2 + ----
          1
          5 + ----
              1
              3 + ----
                  2
    
```

```

class sage.rings.continued_fraction.ContinuedFraction_infinite(w, value=None,
                                                                check=True)
    
```

Bases: *ContinuedFraction_base*

A continued fraction defined by an infinite sequence of partial quotients.

EXAMPLES:

```

sage: t = continued_fraction(words.ThueMorseWord([1,2])); t
↪ needs sage.combinat
[1; 2, 2, 1, 2, 1, 1, 2, 2, 1...]
sage: t.n(digits=100)
    
```

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```

↪needs sage.combinat
1.
↪4223887368827854883415471160245658253068791089917118293118924529164567472725658833124554129620

```

```

>>> from sage.all import *
>>> t = continued_fraction(words.ThueMorseWord([Integer(1), Integer(2)])); t
↪ # needs sage.combinat
[1; 2, 2, 1, 2, 1, 1, 2, 2, 1...]
>>> t.n(digits=Integer(100))
↪ # needs sage.combinat
1.
↪4223887368827854883415471160245658253068791089917118293118924529164567472725658833124554129620

```

We check that comparisons work well:

```

sage: t > continued_fraction(1) and t < continued_fraction(3/2) #_
↪needs sage.combinat
True
sage: t < continued_fraction(1) or t > continued_fraction(2) #_
↪needs sage.combinat
False

```

```

>>> from sage.all import *
>>> t > continued_fraction(Integer(1)) and t < continued_fraction(Integer(3)/
↪Integer(2)) # needs sage.combinat
True
>>> t < continued_fraction(Integer(1)) or t > continued_fraction(Integer(2)) #_
↪ # needs sage.combinat
False

```

Can also be called with a value option:

```

sage: def f(n):
....:     if n % 3 == 2: return 2*(n+1)//3
....:     return 1
sage: w = Word(f, alphabet=NN); w #_
↪needs sage.combinat
word: 1,1,2,1,1,4,1,1,6,1,1,8,1,1,10,1,1,12,1,1,14,1,1,16,1,1,18,1,1,20,1,1,22,1,
↪1,24,1,1,26,1,...
sage: cf = continued_fraction(w, value=e-1); cf #_
↪needs sage.combinat sage.symbolic
[1; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1...]

```

```

>>> from sage.all import *
>>> def f(n):
...     if n % Integer(3) == Integer(2): return Integer(2)*(n+Integer(1))//
↪Integer(3)
...     return Integer(1)
>>> w = Word(f, alphabet=NN); w #_
↪needs sage.combinat
word: 1,1,2,1,1,4,1,1,6,1,1,8,1,1,10,1,1,12,1,1,14,1,1,16,1,1,18,1,1,20,1,1,22,1,
↪1,24,1,1,26,1,...
>>> cf = continued_fraction(w, value=e-Integer(1)); cf #_
↪ # needs sage.combinat sage.symbolic
[1; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1...]

```

In that case a small check is done on the input:

```
sage: cf = continued_fraction(w, value=pi) #_
↳needs sage.combinat sage.symbolic
Traceback (most recent call last):
...
ValueError: value evaluates to 3.141592653589794? while the continued
fraction evaluates to 1.718281828459046? in Real Interval Field
with 53 bits of precision.
```

```
>>> from sage.all import *
>>> cf = continued_fraction(w, value=pi) #_
↳needs sage.combinat sage.symbolic
Traceback (most recent call last):
...
ValueError: value evaluates to 3.141592653589794? while the continued
fraction evaluates to 1.718281828459046? in Real Interval Field
with 53 bits of precision.
```

length()

Return infinity.

EXAMPLES:

```
sage: w = words.FibonacciWord([3,13]) #_
↳needs sage.combinat
sage: cf = continued_fraction(w) #_
↳needs sage.combinat
sage: cf.length() #_
↳needs sage.combinat
+Infinity
```

```
>>> from sage.all import *
>>> w = words.FibonacciWord([Integer(3),Integer(13)]) _
↳ # needs sage.combinat
>>> cf = continued_fraction(w) #_
↳needs sage.combinat
>>> cf.length() #_
↳needs sage.combinat
+Infinity
```

quotient(n)

Return the n-th partial quotient of self.

INPUT:

- n – an integer

EXAMPLES:

```
sage: # needs sage.combinat
sage: w = words.FibonacciWord([1,3])
sage: cf = continued_fraction(w)
sage: cf.quotient(0)
1
sage: cf.quotient(1)
3
sage: cf.quotient(2)
1
```

```
>>> from sage.all import *
>>> # needs sage.combinat
>>> w = words.FibonacciWord([Integer(1), Integer(3)])
>>> cf = continued_fraction(w)
>>> cf.quotient(Integer(0))
1
>>> cf.quotient(Integer(1))
3
>>> cf.quotient(Integer(2))
1
```

quotients()

Return the infinite list from which this continued fraction was built.

EXAMPLES:

```
sage: w = words.FibonacciWord([1,5]) #_
↪needs sage.combinat
sage: cf = continued_fraction(w) #_
↪needs sage.combinat
sage: cf.quotients() #_
↪needs sage.combinat
word: 15115151151151151151151151151151151...
```

```
>>> from sage.all import *
>>> w = words.FibonacciWord([Integer(1), Integer(5)]) #_
↪ # needs sage.combinat
>>> cf = continued_fraction(w) #_
↪needs sage.combinat
>>> cf.quotients() #_
↪needs sage.combinat
word: 15115151151151151151151151151151151...
```

value()

Return the value of self.

If this value was provided on initialization, just return this value otherwise return an element of the real lazy field.

EXAMPLES:

```
sage: def f(n):
....:     if n % 3 == 2: return 2*(n+1)//3
....:     return 1
sage: w = Word(f, alphabet=NN); w #_
↪needs sage.combinat
word: 1,1,2,1,1,4,1,1,6,1,1,8,1,1,10,1,1,12,1,1,14,1,1,16,1,1,18,1,1,20,1,1,
↪22,1,1,24,1,1,26,1,...
sage: cf = continued_fraction(w, value=e-1); cf #_
↪needs sage.combinat sage.symbolic
[1; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1...]
sage: cf.value() #_
↪needs sage.combinat sage.symbolic
e - 1

sage: w = words.FibonacciWord([2,5]) #_
↪needs sage.combinat
```

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```
sage: cf = continued_fraction(w); cf
↳needs sage.combinat
[2; 5, 2, 2, 5, 2, 5, 2, 2, 5, 2, 2, 5, 2, 5, 2, 2, 5, 2, 5...]
sage: cf.value()
↳needs sage.combinat
2.184951302409338?
```

```
>>> from sage.all import *
>>> def f(n):
...     if n % Integer(3) == Integer(2): return Integer(2)*(n+Integer(1))//
↳Integer(3)
...     return Integer(1)
>>> w = Word(f, alphabet=NN); w
↳needs sage.combinat
word: 1,1,2,1,1,4,1,1,6,1,1,8,1,1,10,1,1,12,1,1,14,1,1,16,1,1,18,1,1,20,1,1,
↳22,1,1,24,1,1,26,1,...
>>> cf = continued_fraction(w, value=e-Integer(1)); cf
↳ # needs sage.combinat sage.symbolic
[1; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1...]
>>> cf.value()
↳needs sage.combinat sage.symbolic
e - 1

>>> w = words.FibonacciWord([Integer(2),Integer(5)])
↳ # needs sage.combinat
>>> cf = continued_fraction(w); cf
↳needs sage.combinat
[2; 5, 2, 2, 5, 2, 5, 2, 2, 5, 2, 2, 5, 2, 5, 2, 2, 5, 2, 5...]
>>> cf.value()
↳needs sage.combinat
2.184951302409338?
```

class sage.rings.continued_fraction.**ContinuedFraction_periodic** (*x1*, *x2=None*,
check=True)

Bases: *ContinuedFraction_base*

Continued fraction associated with rational or quadratic number.

A rational number has a finite continued fraction expansion (or ultimately 0). The one of a quadratic number, ie a number of the form $a + b\sqrt{D}$ with a and b rational, is ultimately periodic.

Note: This class stores a tuple `_x1` for the preperiod and a tuple `_x2` for the period. In the purely periodic case `_x1` is empty while in the rational case `_x2` is the tuple `(0,)`.

length()

Return the number of partial quotients of `self`.

EXAMPLES:

```
sage: continued_fraction(2/5).length()
3
sage: cf = continued_fraction([(0,1),(2,)]); cf
[0; 1, (2)*]
sage: cf.length()
+Infinity
```

```

>>> from sage.all import *
>>> continued_fraction(Integer(2)/Integer(5)).length()
3
>>> cf = continued_fraction([(Integer(0), Integer(1)), (Integer(2),)]); cf
[0; 1, (2)*]
>>> cf.length()
+Infinity

```

period()

Return the periodic part of self.

EXAMPLES:

```

sage: # needs sage.rings.number_field
sage: K.<sqrt3> = QuadraticField(3)
sage: cf = continued_fraction(sqrt3); cf
[1; (1, 2)*]
sage: cf.period()
(1, 2)
sage: for k in xrange(2,40):
....:     if not k.is_square():
....:         s = QuadraticField(k).gen()
....:         cf = continued_fraction(s)
....:         print('%2d %d %s' % (k, len(cf.period()), cf))
2 1 [1; (2)*]
3 2 [1; (1, 2)*]
5 1 [2; (4)*]
6 2 [2; (2, 4)*]
7 4 [2; (1, 1, 1, 4)*]
8 2 [2; (1, 4)*]
10 1 [3; (6)*]
11 2 [3; (3, 6)*]
12 2 [3; (2, 6)*]
13 5 [3; (1, 1, 1, 1, 6)*]
14 4 [3; (1, 2, 1, 6)*]
...
35 2 [5; (1, 10)*]
37 1 [6; (12)*]
38 2 [6; (6, 12)*]
39 2 [6; (4, 12)*]

```

```

>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = QuadraticField(Integer(3), names=('sqrt3',)); (sqrt3,) = K._first_
↳ngens(1)
>>> cf = continued_fraction(sqrt3); cf
[1; (1, 2)*]
>>> cf.period()
(1, 2)
>>> for k in xrange(Integer(2), Integer(40)):
...     if not k.is_square():
...         s = QuadraticField(k).gen()
...         cf = continued_fraction(s)
...         print('%2d %d %s' % (k, len(cf.period()), cf))
2 1 [1; (2)*]
3 2 [1; (1, 2)*]
5 1 [2; (4)*]

```

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```

6 2 [2; (2, 4)*]
7 4 [2; (1, 1, 1, 4)*]
8 2 [2; (1, 4)*]
10 1 [3; (6)*]
11 2 [3; (3, 6)*]
12 2 [3; (2, 6)*]
13 5 [3; (1, 1, 1, 1, 6)*]
14 4 [3; (1, 2, 1, 6)*]
...
35 2 [5; (1, 10)*]
37 1 [6; (12)*]
38 2 [6; (6, 12)*]
39 2 [6; (4, 12)*]

```

`period_length()`

Return the number of partial quotients of the preperiodic part of `self`.

EXAMPLES:

```

sage: continued_fraction(2/5).period_length()
1
sage: cf = continued_fraction([(0,1),(2,)]); cf
[0; 1, (2)*]
sage: cf.period_length()
1

```

```

>>> from sage.all import *
>>> continued_fraction(Integer(2)/Integer(5)).period_length()
1
>>> cf = continued_fraction([(Integer(0),Integer(1)),(Integer(2),)]); cf
[0; 1, (2)*]
>>> cf.period_length()
1

```

`preperiod()`

Return the preperiodic part of `self`.

EXAMPLES:

```

sage: # needs sage.rings.number_field
sage: K.<sqrt3> = QuadraticField(3)
sage: cf = continued_fraction(sqrt3); cf
[1; (1, 2)*]
sage: cf.preperiod()
(1,)
sage: cf = continued_fraction(sqrt3/7); cf
[0; 4, (24, 8)*]
sage: cf.preperiod()
(0, 4)

```

```

>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = QuadraticField(Integer(3), names=('sqrt3',)); (sqrt3,) = K._first_
↳ngens(1)
>>> cf = continued_fraction(sqrt3); cf
[1; (1, 2)*]

```

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```
>>> cf.preperiod()
(1,)
>>> cf = continued_fraction(sqrt(3)/Integer(7)); cf
[0; 4, (24, 8)*]
>>> cf.preperiod()
(0, 4)
```

preperiod_length()

Return the number of partial quotients of the preperiodic part of self.

EXAMPLES:

```
sage: continued_fraction(2/5).preperiod_length()
3
sage: cf = continued_fraction([(0,1),(2,)]); cf
[0; 1, (2)*]
sage: cf.preperiod_length()
2
```

```
>>> from sage.all import *
>>> continued_fraction(Integer(2)/Integer(5)).preperiod_length()
3
>>> cf = continued_fraction([(Integer(0),Integer(1)),(Integer(2),)]); cf
[0; 1, (2)*]
>>> cf.preperiod_length()
2
```

quotient(n)

Return the n-th partial quotient of self.

EXAMPLES:

```
sage: cf = continued_fraction([(12,5),(1,3)])
sage: [cf.quotient(i) for i in range(10)]
[12, 5, 1, 3, 1, 3, 1, 3, 1, 3]
```

```
>>> from sage.all import *
>>> cf = continued_fraction([(Integer(12),Integer(5)),(Integer(1),
↳ Integer(3))])
>>> [cf.quotient(i) for i in range(Integer(10))]
[12, 5, 1, 3, 1, 3, 1, 3, 1, 3]
```

value()

Return the value of self as a quadratic number (with square free discriminant).

EXAMPLES:

Some purely periodic examples:

```
sage: cf = continued_fraction([(2,)]); cf
[(2)*]
sage: v = cf.value(); v
↳needs sage.rings.number_field
sqrt(2) + 1
sage: v.continued_fraction()
↳needs sage.rings.number_field
```

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```
[ (2) *]

sage: cf = continued_fraction([()], (1, 2)]); cf
[ (1, 2) *]
sage: v = cf.value(); v                                     #_
↳needs sage.rings.number_field
1/2*sqrt3 + 1/2
sage: v.continued_fraction()                                #_
↳needs sage.rings.number_field
[ (1, 2) *]
```

```
>>> from sage.all import *
>>> cf = continued_fraction([()], (Integer(2),)]); cf
[ (2) *]
>>> v = cf.value(); v                                       #_
↳needs sage.rings.number_field
sqrt2 + 1
>>> v.continued_fraction()                                  #_
↳needs sage.rings.number_field
[ (2) *]

>>> cf = continued_fraction([()], (Integer(1), Integer(2))]); cf
[ (1, 2) *]
>>> v = cf.value(); v                                       #_
↳needs sage.rings.number_field
1/2*sqrt3 + 1/2
>>> v.continued_fraction()                                  #_
↳needs sage.rings.number_field
[ (1, 2) *]
```

The number `sqrt3` that appear above is actually internal to the continued fraction. In order to be access it from the console:

```
sage: cf.value().parent().inject_variables()                 #_
↳needs sage.rings.number_field
Defining sqrt3
sage: sqrt3                                                  #_
↳needs sage.rings.number_field
sqrt3
sage: ((sqrt3+1)/2).continued_fraction()                     #_
↳needs sage.rings.number_field
[ (1, 2) *]
```

```
>>> from sage.all import *
>>> cf.value().parent().inject_variables()                   #_
↳needs sage.rings.number_field
Defining sqrt3
>>> sqrt3                                                    #_
↳needs sage.rings.number_field
sqrt3
>>> ((sqrt3+Integer(1))/Integer(2)).continued_fraction()    _
↳ # needs sage.rings.number_field
[ (1, 2) *]
```

Some ultimately periodic but non periodic examples:

```

sage: cf = continued_fraction([(1,),(2,)]); cf
[1; (2)*]
sage: v = cf.value(); v                                     #_
↪needs sage.rings.number_field
sqrt2
sage: v.continued_fraction()                                #_
↪needs sage.rings.number_field
[1; (2)*]

sage: cf = continued_fraction([(1,3),(1,2)]); cf
[1; 3, (1, 2)*]
sage: v = cf.value(); v                                     #_
↪needs sage.rings.number_field
-sqrt3 + 3
sage: v.continued_fraction()                                #_
↪needs sage.rings.number_field
[1; 3, (1, 2)*]

sage: cf = continued_fraction([(-5,18), (1,3,1,5)])
sage: cf.value().continued_fraction() == cf                #_
↪needs sage.rings.number_field
True
sage: cf = continued_fraction([(-1,),(1,)]))
sage: cf.value().continued_fraction() == cf                #_
↪needs sage.rings.number_field
True

```

```

>>> from sage.all import *
>>> cf = continued_fraction([(Integer(1),),(Integer(2),)]); cf
[1; (2)*]
>>> v = cf.value(); v                                     #_
↪needs sage.rings.number_field
sqrt2
>>> v.continued_fraction()                                #_
↪needs sage.rings.number_field
[1; (2)*]

>>> cf = continued_fraction([(Integer(1),Integer(3)),(Integer(1),
↪Integer(2))]); cf
[1; 3, (1, 2)*]
>>> v = cf.value(); v                                     #_
↪needs sage.rings.number_field
-sqrt3 + 3
>>> v.continued_fraction()                                #_
↪needs sage.rings.number_field
[1; 3, (1, 2)*]

>>> cf = continued_fraction([(-Integer(5),Integer(18)), (Integer(1),
↪Integer(3),Integer(1),Integer(5))])
>>> cf.value().continued_fraction() == cf                #_
↪needs sage.rings.number_field
True
>>> cf = continued_fraction([(-Integer(1),),(Integer(1),)])
>>> cf.value().continued_fraction() == cf                #_
↪needs sage.rings.number_field
True

```

```
class sage.rings.continued_fraction.ContinuedFraction_real(x)
```

Bases: *ContinuedFraction_base*

Continued fraction of a real (exact) number.

This class simply wraps a real number into an attribute (that can be accessed through the method *value()*). The number is assumed to be irrational.

EXAMPLES:

```
sage: cf = continued_fraction(pi); cf                                     #_
↪needs sage.symbolic
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: cf.value()                                                         #_
↪needs sage.symbolic
pi

sage: cf = continued_fraction(e); cf                                     #_
↪needs sage.symbolic
[2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1, ...]
sage: cf.value()                                                         #_
↪needs sage.symbolic
e
```

```
>>> from sage.all import *
>>> cf = continued_fraction(pi); cf                                     #_
↪needs sage.symbolic
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> cf.value()                                                         #_
↪needs sage.symbolic
pi

>>> cf = continued_fraction(e); cf                                     #_
↪needs sage.symbolic
[2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1, ...]
>>> cf.value()                                                         #_
↪needs sage.symbolic
e
```

length()

Return infinity

EXAMPLES:

```
sage: continued_fraction(pi).length()                                   #_
↪needs sage.symbolic
+Infinity
```

```
>>> from sage.all import *
>>> continued_fraction(pi).length()                                     #_
↪needs sage.symbolic
+Infinity
```

quotient(n)

Return the n-th quotient of self.

EXAMPLES:

```

sage: # needs sage.symbolic
sage: cf = continued_fraction(pi)
sage: cf.quotient(27)
13
sage: cf.quotient(2552)
152
sage: cf.quotient(10000)          # long time
5

```

```

>>> from sage.all import *
>>> # needs sage.symbolic
>>> cf = continued_fraction(pi)
>>> cf.quotient(Integer(27))
13
>>> cf.quotient(Integer(2552))
152
>>> cf.quotient(Integer(10000))    # long time
5

```

The algorithm is not efficient with element of the symbolic ring and, if possible, one can always prefer number fields elements. The reason is that, given a symbolic element x , there is no automatic way to evaluate in RIF an expression of the form $(a*x+b)/(c*x+d)$ where both the numerator and the denominator are extremely small:

```

sage: # needs sage.symbolic
sage: a1 = pi
sage: c1 = continued_fraction(a1)
sage: p0 = c1.numerator(12); q0 = c1.denominator(12)
sage: p1 = c1.numerator(13); q1 = c1.denominator(13)
sage: num = (q0*a1 - p0); num.n()
1.49011611938477e-8
sage: den = (q1*a1 - p1); den.n()
-2.98023223876953e-8
sage: a1 = -num/den
sage: RIF(a1)
[-infinity .. +infinity]

```

```

>>> from sage.all import *
>>> # needs sage.symbolic
>>> a1 = pi
>>> c1 = continued_fraction(a1)
>>> p0 = c1.numerator(Integer(12)); q0 = c1.denominator(Integer(12))
>>> p1 = c1.numerator(Integer(13)); q1 = c1.denominator(Integer(13))
>>> num = (q0*a1 - p0); num.n()
1.49011611938477e-8
>>> den = (q1*a1 - p1); den.n()
-2.98023223876953e-8
>>> a1 = -num/den
>>> RIF(a1)
[-infinity .. +infinity]

```

The same computation with an element of a number field instead of π gives a very satisfactory answer:

```

sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a2> = NumberField(x^3 - 2, embedding=1.25)

```

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```
sage: c2 = continued_fraction(a2)
sage: p0 = c2.numerator(111); q0 = c2.denominator(111)
sage: p1 = c2.numerator(112); q1 = c2.denominator(112)
sage: num = (q0*a2 - p0); num.n()
-4.56719261665907e46
sage: den = (q1*a2 - p1); den.n()
-3.65375409332726e47
sage: a2 = -num/den
sage: b2 = RIF(a2); b2
1.002685823312715?
sage: b2.absolute_diameter()
8.88178419700125e-16
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(2), embedding=RealNumber('1.25'),
↳ names=('a2',)); (a2,) = K._first_ngens(1)
>>> c2 = continued_fraction(a2)
>>> p0 = c2.numerator(Integer(111)); q0 = c2.denominator(Integer(111))
>>> p1 = c2.numerator(Integer(112)); q1 = c2.denominator(Integer(112))
>>> num = (q0*a2 - p0); num.n()
-4.56719261665907e46
>>> den = (q1*a2 - p1); den.n()
-3.65375409332726e47
>>> a2 = -num/den
>>> b2 = RIF(a2); b2
1.002685823312715?
>>> b2.absolute_diameter()
8.88178419700125e-16
```

The consequence is that the precision needed with `c1` grows when we compute larger and larger partial quotients:

```
sage: # needs sage.symbolic
sage: c1.quotient(100)
2
sage: c1._xa.parent()
Real Interval Field with 353 bits of precision
sage: c1.quotient(200)
3
sage: c1._xa.parent()
Real Interval Field with 753 bits of precision
sage: c1.quotient(300)
5
sage: c1._xa.parent()
Real Interval Field with 1053 bits of precision

sage: # needs sage.rings.number_field
sage: c2.quotient(200)
6
sage: c2._xa.parent()
Real Interval Field with 53 bits of precision
sage: c2.quotient(500)
1
sage: c2._xa.parent()
```

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```
Real Interval Field with 53 bits of precision
sage: c2.quotient(1000)
1
sage: c2._xa.parent()
Real Interval Field with 53 bits of precision
```

```
>>> from sage.all import *
>>> # needs sage.symbolic
>>> c1.quotient(Integer(100))
2
>>> c1._xa.parent()
Real Interval Field with 353 bits of precision
>>> c1.quotient(Integer(200))
3
>>> c1._xa.parent()
Real Interval Field with 753 bits of precision
>>> c1.quotient(Integer(300))
5
>>> c1._xa.parent()
Real Interval Field with 1053 bits of precision

>>> # needs sage.rings.number_field
>>> c2.quotient(Integer(200))
6
>>> c2._xa.parent()
Real Interval Field with 53 bits of precision
>>> c2.quotient(Integer(500))
1
>>> c2._xa.parent()
Real Interval Field with 53 bits of precision
>>> c2.quotient(Integer(1000))
1
>>> c2._xa.parent()
Real Interval Field with 53 bits of precision
```

value()

Return the value of self (the number from which it was built).

EXAMPLES:

```
sage: cf = continued_fraction(e) #_
↪needs sage.symbolic
sage: cf.value() #_
↪needs sage.symbolic
e
```

```
>>> from sage.all import *
>>> cf = continued_fraction(e) #_
↪needs sage.symbolic
>>> cf.value() #_
↪needs sage.symbolic
e
```

`sage.rings.continued_fraction.check_and_reduce_pair` ($x1$, $x2=None$)

There are often two ways to represent a given continued fraction. This function makes it canonical.

In the very special case of the number 0 we return the pair $((0,), (0,))$.

`sage.rings.continued_fraction.continued_fraction(x, value=None)`

Return the continued fraction of x .

INPUT:

- x – a number or a list of partial quotients (for finite development) or two list of partial quotients (preperiod and period for ultimately periodic development)

EXAMPLES:

A finite continued fraction may be initialized by a number or by its list of partial quotients:

```
sage: continued_fraction(12/571)
[0; 47, 1, 1, 2, 2]
sage: continued_fraction([3, 2, 1, 4])
[3; 2, 1, 4]
```

```
>>> from sage.all import *
>>> continued_fraction(Integer(12)/Integer(571))
[0; 47, 1, 1, 2, 2]
>>> continued_fraction([Integer(3), Integer(2), Integer(1), Integer(4)])
[3; 2, 1, 4]
```

It can be called with elements defined from symbolic values, in which case the partial quotients are evaluated in a lazy way:

```
sage: c = continued_fraction(golden_ratio); c #_
↳needs sage.symbolic
[1; 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...]
sage: c.convergent(12) #_
↳needs sage.symbolic
377/233
sage: fibonacci(14)/fibonacci(13) #_
↳needs sage.libs.pari
377/233

sage: # needs sage.symbolic
sage: continued_fraction(pi)
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
sage: a = c.convergent(3); a
355/113
sage: a.n()
3.14159292035398
sage: pi.n()
3.14159265358979

sage: # needs sage.symbolic
sage: continued_fraction(sqrt(2))
[1; 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, ...]
sage: continued_fraction(tan(1))
[1; 1, 1, 3, 1, 5, 1, 7, 1, 9, 1, 11, 1, 13, 1, 15, 1, 17, 1, 19, ...]
sage: continued_fraction(tanh(1))
[0; 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, ...]
sage: continued_fraction(e)
[2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1, ...]
```

```

>>> from sage.all import *
>>> c = continued_fraction(golden_ratio); c                                     #_
↳needs sage.symbolic
[1; 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...]
>>> c.convergent(Integer(12))                                                  _
↳ # needs sage.symbolic
377/233
>>> fibonacci(Integer(14))/fibonacci(Integer(13))                             _
↳ # needs sage.libs.pari
377/233

>>> # needs sage.symbolic
>>> continued_fraction(pi)
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> c = continued_fraction(pi); c
[3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, ...]
>>> a = c.convergent(Integer(3)); a
355/113
>>> a.n()
3.14159292035398
>>> pi.n()
3.14159265358979

>>> # needs sage.symbolic
>>> continued_fraction(sqrt(Integer(2)))
[1; 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, ...]
>>> continued_fraction(tan(Integer(1)))
[1; 1, 1, 3, 1, 5, 1, 7, 1, 9, 1, 11, 1, 13, 1, 15, 1, 17, 1, 19, ...]
>>> continued_fraction(tanh(Integer(1)))
[0; 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, ...]
>>> continued_fraction(e)
[2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 1, 1, 12, 1, 1, ...]

```

If you want to play with quadratic numbers (such as `golden_ratio` and `sqrt(2)` above), it is much more convenient to use number fields as follows since preperiods and periods are computed:

```

sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<sqrt5> = NumberField(x^2 - 5, embedding=2.23)
sage: my_golden_ratio = (1 + sqrt5)/2
sage: cf = continued_fraction((1+sqrt5)/2); cf
[(1)*]
sage: cf.convergent(12)
377/233
sage: cf.period()
(1,)
sage: cf = continued_fraction(2/3+sqrt5/5); cf
[1; 8, (1, 3, 1, 1, 3, 9)*]
sage: cf.preperiod()
(1, 8)
sage: cf.period()
(1, 3, 1, 1, 3, 9)

sage: # needs sage.rings.number_field
sage: L.<sqrt2> = NumberField(x^2 - 2, embedding=1.41)
sage: cf = continued_fraction(sqrt2); cf
[1; (2)*]

```

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```
sage: cf.period()
(2,)
sage: cf = continued_fraction(sqrt(2)/3); cf
[0; 2, (8, 4)*]
sage: cf.period()
(8, 4)
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(2) - Integer(5), embedding=RealNumber('2.23'),
↳names=('sqrt5',)); (sqrt5,) = K._first_ngens(1)
>>> my_golden_ratio = (Integer(1) + sqrt5)/Integer(2)
>>> cf = continued_fraction((Integer(1)+sqrt5)/Integer(2)); cf
[(1)*]
>>> cf.convergent(Integer(12))
377/233
>>> cf.period()
(1,)
>>> cf = continued_fraction(Integer(2)/Integer(3)+sqrt5/Integer(5)); cf
[1; 8, (1, 3, 1, 1, 3, 9)*]
>>> cf.preperiod()
(1, 8)
>>> cf.period()
(1, 3, 1, 1, 3, 9)

>>> # needs sage.rings.number_field
>>> L = NumberField(x**Integer(2) - Integer(2), embedding=RealNumber('1.41'),
↳names=('sqrt2',)); (sqrt2,) = L._first_ngens(1)
>>> cf = continued_fraction(sqrt2); cf
[1; (2)*]
>>> cf.period()
(2,)
>>> cf = continued_fraction(sqrt2/Integer(3)); cf
[0; 2, (8, 4)*]
>>> cf.period()
(8, 4)
```

It is also possible to go the other way around, build a ultimately periodic continued fraction from its preperiod and its period and get its value back:

```
sage: cf = continued_fraction([(1,1), (2,8)]); cf
[1; 1, (2, 8)*]
sage: cf.value()
↳needs sage.rings.number_field #_
2/11*sqrt5 + 14/11
```

```
>>> from sage.all import *
>>> cf = continued_fraction([(Integer(1),Integer(1)), (Integer(2),Integer(8))]);
↳cf
[1; 1, (2, 8)*]
>>> cf.value()
↳needs sage.rings.number_field #_
2/11*sqrt5 + 14/11
```

It is possible to deal with higher degree number fields but in that case the continued fraction expansion is known to

be aperiodic:

```
sage: K.<a> = NumberField(x^3 - 2, embedding=1.25) #_
↳needs sage.rings.number_field
sage: cf = continued_fraction(a); cf #_
↳needs sage.rings.number_field
[1; 3, 1, 5, 1, 1, 4, 1, 1, 8, 1, 14, 1, 10, 2, 1, 4, 12, 2, 3, ...]
```

```
>>> from sage.all import *
>>> K = NumberField(x**Integer(3) - Integer(2), embedding=RealNumber('1.25'), #_
↳names=('a',)); (a,) = K._first_ngens(1) # needs sage.rings.number_field
>>> cf = continued_fraction(a); cf #_
↳needs sage.rings.number_field
[1; 3, 1, 5, 1, 1, 4, 1, 1, 8, 1, 14, 1, 10, 2, 1, 4, 12, 2, 3, ...]
```

Note that initial rounding can result in incorrect trailing partial quotients:

```
sage: continued_fraction(RealField(39)(e)) #_
↳needs sage.symbolic
[2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 2]
```

```
>>> from sage.all import *
>>> continued_fraction(RealField(Integer(39))(e)) #_
↳ # needs sage.symbolic
[2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, 1, 1, 10, 2]
```

Note the value returned for floating point number is the continued fraction associated to the rational number you obtain with a conversion:

```
sage: for _ in range(10):
.....:     x = RR.random_element()
.....:     cff = continued_fraction(x)
.....:     cfe = QQ(x).continued_fraction()
.....:     assert cff == cfe, "%s %s %s"%(x,cff,cfe)
```

```
>>> from sage.all import *
>>> for _ in range(Integer(10)):
...     x = RR.random_element()
...     cff = continued_fraction(x)
...     cfe = QQ(x).continued_fraction()
...     assert cff == cfe, "%s %s %s"%(x,cff,cfe)
```

`sage.rings.continued_fraction.continued_fraction_list` (*x*, *type='std'*,
partial_convergents=False,
bits=None, *nterms=None*)

Return the (finite) continued fraction of *x* as a list.

The continued fraction expansion of *x* are the coefficients a_i in

$$x = a_0 + 1/(a_1 + 1/(...))$$

with a_0 integer and $a_1, \dots$ positive integers. The Hirzebruch-Jung continued fraction is the one for which the + signs are replaced with − signs

$$x = a_0 - 1/(a_1 - 1/(...))$$

See also:

`continued_fraction()`

INPUT:

- `x` – exact rational or floating-point number. The number to compute the continued fraction of.
- `type` – either “std” (default) for standard continued fractions or “hj” for Hirzebruch-Jung ones.
- `partial_convergents` – boolean. Whether to return the partial convergents.
- `bits` – an optional integer that specifies a precision for the real interval field that is used internally.
- `nterms` – integer. The upper bound on the number of terms in the continued fraction expansion to return.

OUTPUT:

A list of integers, the coefficients in the continued fraction expansion of `x`. If `partial_convergents` is set to True, then return a pair containing the coefficient list and the partial convergents list is returned.

EXAMPLES:

```
sage: continued_fraction_list(45/19)
[2, 2, 1, 2, 2]
sage: 2 + 1/(2 + 1/(1 + 1/(2 + 1/2)))
45/19

sage: continued_fraction_list(45/19, type="hj")
[3, 2, 3, 2, 3]
sage: 3 - 1/(2 - 1/(3 - 1/(2 - 1/3)))
45/19
```

```
>>> from sage.all import *
>>> continued_fraction_list(Integer(45)/Integer(19))
[2, 2, 1, 2, 2]
>>> Integer(2) + Integer(1)/(Integer(2) + Integer(1)/(Integer(1) +
Integer(1)/(Integer(2) + Integer(1)/Integer(2))))
45/19

>>> continued_fraction_list(Integer(45)/Integer(19), type="hj")
[3, 2, 3, 2, 3]
>>> Integer(3) - Integer(1)/(Integer(2) - Integer(1)/(Integer(3) -
Integer(1)/(Integer(2) - Integer(1)/Integer(3))))
45/19
```

Specifying `bits` or `nterms` modify the length of the output:

```
sage: # needs sage.symbolic
sage: continued_fraction_list(e, bits=20)
[2, 1, 2, 1, 1, 4, 2]
sage: continued_fraction_list(sqrt(2) + sqrt(3), bits=30)
[3, 6, 1, 5, 7, 2]
sage: continued_fraction_list(pi, bits=53)
[3, 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14]
sage: continued_fraction_list(log(3/2), nterms=15)
[0, 2, 2, 6, 1, 11, 2, 1, 2, 2, 1, 4, 3, 1, 1]
sage: continued_fraction_list(tan(sqrt(pi)), nterms=20)
[-5, 9, 4, 1, 1, 1, 1, 1, 1, 5, 1, 1, 1, 1, 1, 2, 4, 3, 1, 63]
```

```

>>> from sage.all import *
>>> # needs sage.symbolic
>>> continued_fraction_list(e, bits=Integer(20))
[2, 1, 2, 1, 1, 4, 2]
>>> continued_fraction_list(sqrt(Integer(2)) + sqrt(Integer(3)), bits=Integer(30))
[3, 6, 1, 5, 7, 2]
>>> continued_fraction_list(pi, bits=Integer(53))
[3, 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14]
>>> continued_fraction_list(log(Integer(3)/Integer(2)), nterms=Integer(15))
[0, 2, 2, 6, 1, 11, 2, 1, 2, 2, 1, 4, 3, 1, 1]
>>> continued_fraction_list(tan(sqrt(pi)), nterms=Integer(20))
[-5, 9, 4, 1, 1, 1, 1, 1, 1, 5, 1, 1, 1, 1, 1, 2, 4, 3, 1, 63]

```

When the continued fraction is infinite (ie x is an irrational number) and the parameters `bits` and `nterms` are not specified then a warning is raised:

```

sage: continued_fraction_list(sqrt(2)) #_
↳ needs sage.symbolic
doctest:...: UserWarning: the continued fraction of sqrt(2) seems infinite,
return only the first 20 terms
[1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2]
sage: continued_fraction_list(sqrt(4/19)) #_
↳ needs sage.symbolic
doctest:...: UserWarning: the continued fraction of 2*sqrt(1/19) seems infinite,
return only the first 20 terms
[0, 2, 5, 1, 1, 2, 1, 16, 1, 2, 1, 1, 5, 4, 5, 1, 1, 2, 1, 16]

```

```

>>> from sage.all import *
>>> continued_fraction_list(sqrt(Integer(2))) #_
↳ # needs sage.symbolic
doctest:...: UserWarning: the continued fraction of sqrt(2) seems infinite,
return only the first 20 terms
[1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2]
>>> continued_fraction_list(sqrt(Integer(4)/Integer(19))) #_
↳ # needs sage.symbolic
doctest:...: UserWarning: the continued fraction of 2*sqrt(1/19) seems infinite,
return only the first 20 terms
[0, 2, 5, 1, 1, 2, 1, 16, 1, 2, 1, 1, 5, 4, 5, 1, 1, 2, 1, 16]

```

An examples with the list of partial convergents:

```

sage: continued_fraction_list(RR(pi), partial_convergents=True) #_
↳ needs sage.symbolic
([3, 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 3],
 [(3, 1),
  (22, 7),
  (333, 106),
  (355, 113),
  (103993, 33102),
  (104348, 33215),
  (208341, 66317),
  (312689, 99532),
  (833719, 265381),
  (1146408, 364913),
  (4272943, 1360120),
  (5419351, 1725033),
  (80143857, 25510582),

```

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```
(245850922, 78256779)])
```

```
>>> from sage.all import *
>>> continued_fraction_list(RR(pi), partial_convergents=True) #_
↳needs sage.symbolic
([3, 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 3],
 [(3, 1),
  (22, 7),
  (333, 106),
  (355, 113),
  (103993, 33102),
  (104348, 33215),
  (208341, 66317),
  (312689, 99532),
  (833719, 265381),
  (1146408, 364913),
  (4272943, 1360120),
  (5419351, 1725033),
  (80143857, 25510582),
  (245850922, 78256779)])
```

`sage.rings.continued_fraction.convergents(x)`

Return the (partial) convergents of the number x .

EXAMPLES:

```
sage: from sage.rings.continued_fraction import convergents
sage: convergents(143/255)
[0, 1, 1/2, 4/7, 5/9, 9/16, 14/25, 23/41, 60/107, 143/255]
```

```
>>> from sage.all import *
>>> from sage.rings.continued_fraction import convergents
>>> convergents(Integer(143)/Integer(255))
[0, 1, 1/2, 4/7, 5/9, 9/16, 14/25, 23/41, 60/107, 143/255]
```

`sage.rings.continued_fraction.last_two_convergents(x)`

Given the list x that consists of numbers, return the two last convergents $p_{n-1}, q_{n-1}, p_n, q_n$.

This function is principally used to compute the value of a ultimately periodic continued fraction.

OUTPUT: a 4-tuple of Sage integers

EXAMPLES:

```
sage: from sage.rings.continued_fraction import last_two_convergents
sage: last_two_convergents([])
(0, 1, 1, 0)
sage: last_two_convergents([0])
(1, 0, 0, 1)
sage: last_two_convergents([-1, 1, 3, 2])
(-1, 4, -2, 9)
```

```
>>> from sage.all import *
>>> from sage.rings.continued_fraction import last_two_convergents
>>> last_two_convergents([])
(0, 1, 1, 0)
```

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```
>>> last_two_convergents([Integer(0)])
(1, 0, 0, 1)
>>> last_two_convergents([-Integer(1), Integer(1), Integer(3), Integer(2)])
(-1, 4, -2, 9)
```

`sage.rings.continued_fraction.rat_interval_cf_list(r1, r2)`

Return the common prefix of the rationals `r1` and `r2` seen as continued fractions.

OUTPUT: a list of Sage integers.

EXAMPLES:

```
sage: from sage.rings.continued_fraction import rat_interval_cf_list
sage: rat_interval_cf_list(257/113, 5224/2297)
[2, 3, 1, 1, 1, 4]
sage: for prec in range(10, 54):
↳needs sage.rings.real_interval_field
....:     R = RealIntervalField(prec)
....:     for _ in range(100):
....:         x = R.random_element() * R.random_element() + R.random_element() /
↳100
....:         l = x.lower().exact_rational()
....:         u = x.upper().exact_rational()
....:         if l.floor() != u.floor():
....:             continue
....:         cf = rat_interval_cf_list(l, u)
....:         a = continued_fraction(cf).value()
....:         b = continued_fraction(cf+[1]).value()
....:         if a > b:
....:             a, b = b, a
....:         assert a <= l
....:         assert b >= u
```

```
>>> from sage.all import *
>>> from sage.rings.continued_fraction import rat_interval_cf_list
>>> rat_interval_cf_list(Integer(257)/Integer(113), Integer(5224)/Integer(2297))
[2, 3, 1, 1, 1, 4]
>>> for prec in range(Integer(10), Integer(54)):
↳# needs sage.rings.real_interval_field
...     R = RealIntervalField(prec)
...     for _ in range(Integer(100)):
...         x = R.random_element() * R.random_element() + R.random_element() /
↳Integer(100)
...         l = x.lower().exact_rational()
...         u = x.upper().exact_rational()
...         if l.floor() != u.floor():
...             continue
...         cf = rat_interval_cf_list(l, u)
...         a = continued_fraction(cf).value()
...         b = continued_fraction(cf+[Integer(1)]).value()
...         if a > b:
...             a, b = b, a
...         assert a <= l
...         assert b >= u
```


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